

Transmission of the Quantum Interaction of Erythrocytes

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Abstract

The previously demonstrated long-range quantum mechanical interaction between human erythrocytes suspended in plasma also occurs with artificial media. Certain macromolecules dissolved in phosphate-buffered saline will transmit the interaction provided their concentration is above a minimum. Extended macromolecules transmit the interaction, whereas a compact macromolecule (albumin) does not.

Index Entries: Erythrocytes, quantum interaction between; quantum mechanics, and erythrocyte interactions; macromolecules, and quantum erythrocyte interactions; ultra-long-range force, between erythrocytes.

Introduction

We have shown that human erythrocytes interact when less than a few micrometers apart (1-3) and that this interaction has features in common with Fröhlich's (4, 5) theory of ultra-long-range coherent quantum mechanical interactions in biological systems. The electric field of the membrane potential is about ten million volts per meter, which polarizes the molecules constituting the membrane. The polarized molecules interact, giving rise to modes of vibration that Fröhlich (5) calculates to have frequencies of the order of 10^{11} Hz, at which frequency quantum mechanical properties become significant. In the very strong electric field, conformational change in membrane molecules and the readjustment of counterions will give rise

to metastable states of polarization that are analogous to the states of "population inversion" in the photon laser, though within the membrane the quantum particles are phonons. The theory asserts that if the membrane is supplied with metabolic energy above a minimum level, membrane phonons will trigger into a laser-like coherent mode of vibration that can give rise to forces between membranes, organelles, and macromolecules extending over a range of several micrometers (5).

We made measurements of the Brownian movement of human erythrocytes in blood that showed that the movement is no longer random when two erythrocytes are closer than three cell radii (2, 3). The interaction disappears when the cells are killed and it disappears *reversibly* when the membrane potential is abolished or when the cells are metabolically depleted. Moreover, there is some evidence from experiments on interspecies erythrocyte rouleau formation (6) that the interaction is specific (2, 3). At low frequencies, ionic screening would limit the range of a polar-wave interaction to a few nanometers, but Fröhlich demonstrates (5) that at 10^{11} Hz ionic screening is insignificant up to distances of micrometers. Pohl (7, 8) has shown that dividing cells will attract BaTiO_3 particles (dielectric constant, 4000) more readily than BaSO_4 particles (dielectric constant 11). He names the process "microdielectrophoresis" (9), and attributes the interaction to cellular electromagnetic vibrations of the Fröhlich type, but at a lower frequency. With erythrocytes (murine) he found no interaction. These experiments of Pohl (7) were done with cells suspended in artificial media of very low salt concentration, which enhances electromagnetic transmission. Our experiments were done with the erythrocytes suspended in their own plasma. When the cells were suspended in Ringer-Locke solution or phosphate-buffered saline (high salt content similar to that of plasma) there was no interaction. In the experiments described in this paper, we have tested the hypothesis that macromolecules are necessary to transmit the quantum interaction of human erythrocytes. The implication is that this interaction is not a direct polar wave attraction of the type demonstrated by Pohl (7, 8).

Methods

The methods and the Brownian theory (10) employed in the calculations have been fully described (3). The observed process is Brownian movement of erythrocytes, on which the Fröhlich interaction is superimposed. Provided there is an adequate concentration of certain macromolecules in the suspending medium, red cells that come into contact form themselves into easily recognized columns of cells called rouleaux. The macromolecules under study are polyvinylpyrrolidone (PVP-360, average molecular weight 360,000; Sigma Chemicals Co., St. Louis, MO), dextran (Dextran 70, molecular weight 60,000–90,000, ICM, Cleveland, OH), human fibrinogen prepared by the Cohn fractionation method, and human serum albumin (Connaught Laboratories, Willowdale, Ontario). With the exception of albumin, the macromolecules are known rouleau-inducing agents. The macromolecules were made into solutions of various concentrations in phosphate-buffered saline (PBS: composition 150 mM NaCl, 7.3 mM Na_2HPO_4 , 2.8 mM

NaH_2PO_4 , 2.2 mM CaCl_2 , 4.6 mM MgCl_2 , 310 mosmol, pH 7.4 at 22°C). The viscosity of the solutions was measured with a cone and plate viscometer (Synchro-Lectric, Brookfield, Stoughton, MA). At the start of an experiment the solution was filtered through a 0.3 μm Millipore filter and human blood from a finger prick was added to give a cell concentration (v/v) of $\sim 0.2\%$. The suspension was stirred and quickly transferred to a siliconized (Dow Corning, Midland, MI: Sylgard 184 or R4-3117) hemacytometer chamber (Spencer Bright-Line, American Optical Corp., Buffalo, NY). The chamber was sealed with silicone grease to prevent evaporation and photographed with a time-lapse ciné-camera on an Ortholux microscope (Wild-Leitz, Rockleigh, NJ) at about 60 $\times$ magnification. An exposure was taken every 15 s for 1 h.

Analysis of Results

The observed process is an aggregation of cells undergoing Brownian movement on which is superimposed a specific interaction at a distance. The cells sediment to the floor of the chamber and if the motion is purely Brownian the kinetics of rouleau-formation is given (3, 10) by

$$\frac{1}{N_t} - \frac{1}{N_o} = \frac{2kT}{\eta} t \quad [1]$$

where k is Boltzmann's constant, T is the absolute temperature, and η is the viscosity of the solution. The developed ciné-film was projected with a Photo-optical Data Analyser (Model 224A, L-W International, Woodland Hills, CA) and in the earliest frame at which the cells had settled the total number of cells and rouleaux (N_o) was counted in an area defined by the hemacytometer grid. The count was repeated (N_t) every fifth frame (75 s) and the regression of N_t^{-1} on t was computed. Correlation coefficients averaged 0.98.

Equation [1] is derived on the assumption that the diffusion coefficient is given by the Stokes-Einstein relation

$$D = \frac{kT}{6\pi\eta r} \quad [2]$$

where the denominator is the Stokes' drag coefficient

$$\mu = 6\pi\eta r \quad [3]$$

We have done an extensive series of measurements on the drag coefficients under the conditions of our experiments (2, 3, 11). These demonstrate (3) that, in plasma, the experimental value is 1.78 ± 0.06 (SEM) times greater than the theoretical value, and they imply that cells in contact with a surface experience an added hydrodynamic drag (11). Accordingly, the rate constant in Eq. [1] has to be divided by the ratio of the experimentally determined to the theoretical drag coefficient. The drag coefficient is measured in each experiment from the Brownian movement of isolated cells (2, 3) and for each experiment a Brownian motion rate

constant is calculated. This is the factor $2kT/\eta$ in Eq. [1] corrected for the additional hydrodynamic drag. For a viscosity of 1.7×10^{-3} Pas, this rate constant has a value of $2.6 \times 10^{-18} \text{m}^3 \text{s}^{-1}$. For convenience we use an interaction coefficient Ξ which we define to have a value of unity when representing the Brownian motion rate constant. Experiments on dead cells and microspheres (2, 3) demonstrate values for Ξ that do not differ significantly from unity. It is shown in reference (3) that if r is the erythrocyte radius, then $Z = \Xi r$ where Z is the radius of the zone of attraction around the erythrocyte.

Results

The results are given in Table 1. Concentrations of macromolecules are given in column 1 and the viscosities of the suspending media in column 2. The measured drag coefficients in column 3 are used to calculate the Brownian motion rate constant as explained under Methods. The interaction coefficients (Ξ column 4) are expressed as multiples of this Brownian motion rate constant. If $\Xi > 1$ an attractive interaction is implied and $\Xi < 1$ implies a repulsion. It will be seen that with

TABLE 1
Results

Concentration of macromolecule, g/100 mL	Viscosity of suspension, $\eta \times 10^3$ Pas	Drag coefficient $\mu \times 10^7 \text{ Nsm}^{-1}$	Interaction coefficient, Ξ	No. of experiments
Polyvinylpyrrolidone				
0.05	1.06	3.05	0.67	2
0.1	1.18	3.30	1.75	2
0.2	1.39	2.47	2.92	2
0.4	1.85	3.01	1.94	2
0.6	2.23	4.46	2.98	2
Dextran				
<1.0	1 to 1.13	2.75 ± 0.60^a	0.23 ± 0.11^a	12
2.0	1.51	2.38	1.66	2
4.0	2.11	2.84	1.60	2
6.0	2.87	5.27	3.10	2
Fibrinogen				
0.1	1.70	3.98	0.16	1
0.3	1.97	2.50	3.74	2
0.4	2.00	2.42	4.35	2
Albumin				
4.0	1.24	2.48	0.10	2
6.0	1.45	3.73	0.18	2
8.0	1.63	3.08	0.18	2

^aStandard deviation.

PVP concentration ≥ 0.1 g%, Dextran ≥ 2.0 g%, and fibrinogen ≥ 0.3 g%, Ξ is greater than one. The concentrations of fibrinogen that showed an interaction were in the physiological range, whereas albumin at physiological concentrations gave no indication of an attraction.

In the previously described experiments (3) that were all done in plasma, rouleaux formed even when the cells were metabolically depleted or with a zero membrane potential. The counts of (N_o) and (N_i) were unequivocal, although the magnification and resolution used was low to get sufficient cells in the field of view for statistical analysis. With low concentrations of PVP, dextran, and fibrinogen, and with all concentrations of albumin, rouleaux do not form, but a few cells stick together. These are identified by running the film quickly back and forth to check that they indeed remain together though human error may occur because the counting process is long and tedious. For instance, we did pairs of experiments at six different concentrations of dextran between 0.1 and 1.0 g%, but the values of Ξ were distributed randomly. Hence the presentation of these results in the table as a mean and standard deviation. It is possible that there is a threshold concentration above which the quantum interaction appears as a step function. This hypothesis could only be tested by many more experiments, work that will not be done until automated methods of film analysis are available.

The drag coefficients should, of course, change uniformly as the viscosity and the density of the suspending medium increases, but this was not so. In early experiments with plasma we encountered difficulty in getting uniform drag coefficients, but the problem was overcome (3, 11). Further work on surface interaction when employing solutions other than plasma should improve the accuracy of the results. Nevertheless, it is important to remember that the drag coefficient was *measured* in each experiment and that the interaction coefficient is a measure of any deviation from the *observed* Brownian motion of isolated cells.

Discussion

Rowlands et al. (1-3) have shown that normal human red cells in their own plasma form rouleaux significantly faster than Brownian motion predicts. Varying the membrane potential and the cellular ATP content shows that the interaction is compatible with the major predictions of Fröhlich's theory (4, 5) of coherent excitations in biological systems. Cells suspended in phosphate-buffered saline do not interact. The present results show that extended macromolecules must be present in sufficient concentration in the suspending medium, a point that was not predicted by Fröhlich's theory in its present form. PVP and dextran are electrically neutral and fibrinogen carries a net negative charge. Albumin, a compact globular molecule, does not transmit the interaction even when its concentration is at or above the upper limit of normal in plasma. This suggests that the secondary or tertiary structure of the macromolecule determines whether or not it will transmit the Fröhlich interaction.

Human erythrocytes, in addition to having a membrane potential, carry a negative surface charge (12, 13) that keeps the membranes apart when the cells are sus-

pended in isotonic solutions. The surface charge is neutralized by counterions up to very short distances of approach, quite invisible in our microscope, of necessity used at low magnification. Brownian movement can overcome a small repulsion and two such erythrocytes may stick, but if they do not, Brownian motion will cause them to separate. This process is monitored by running the ciné-film back and forth (see Results). Nevertheless, with the experiments performed as at present, firm conclusions cannot be drawn from the magnitude of values of the interaction coefficient which are substantially less than unity as exemplified by the results for low concentrations of dextran.

In the plasma experiments (2, 3) there was a considerable spread in the values for Ξ for normal cells. This is not unexpected. There is a considerable spread in the area and volume of the erythrocytes in a single sample of human blood (14) and an even larger spread in the membrane permeability for individual cells to certain nonelectrolytes (15). Therefore, the Fröhlich force is unlikely to be identical between every pair of erythrocytes. And so the variations in the values of Ξ for PVP ≥ 0.1 g% probably reflect the variations noted in the plasma experiments (3).

In spite of the reservations discussed above, it can be concluded that the quantum mechanical interaction between human erythrocytes can only be transmitted in the presence of a sufficient concentration of extended macromolecules. Pohl (7, 8) has demonstrated an attractive force toward mitotic cells in suspending media devoid of macromolecules, but only if the medium has a very low ionic strength. He did not find an effect with murine erythrocytes.

Our experiments were done at physiological ionic strengths and indeed the original experiments were done in plasma itself (1–3), so there is no incompatibility between Pohl's findings and our own. Fröhlich has reviewed (5) the evidence for the occurrence of vibrations of the predicted frequency in microorganisms and in cells in tissue culture. Pohl's work and ours demonstrate the predicted ultra-long-range interaction and so all the major features of Fröhlich's theory of coherent excitations in biological systems have been confirmed.

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